

ASPIRE · TECH/BIO workflow

YAML configuration · Nextflow / Mamba · resumable execution · audited outputs

Reads → ASVs → full taxonomy

Shared QC for biological samples, BIO controls and TECH blanks

Optional Cutadapt → fastp / VSEARCH → ASV count matrix → SINA / QIIME 2 taxonomy



Optional TECH / BIO control decontamination

≥5,000 post-QC reads for biological samples; retain all nonzero controls

Independent prevalence tests → union ASV removal; retained counts unchanged

Positive controls stay in QC. Either control arm may be disabled.



Reference screening → FILTER_ASVS

Host / mitochondrial evidence → biological abundance and taxonomy filters

Mock: ≥0.1% RA in any biological sample AND nonzero counts in ≥5% of biological samples

This percentage is separate from decontam's prevalence-score cutoffs.



Metadata → selected analysis tables

PLOT_METADATA receives the final filtered biological count table

Optional cohort selection for other analyses; diversity retains all final biological samples



Optional analyses and integration

Diversity · indicators · patient contrasts · power · participant-level ASV–VOC tests

SPIEC-EASI networks / topology / modules · filtered ASV-to-genome linkage



Results, QC and provenance

Tables · editable figures · removal audits · HTML report · logs and checksums

CAMI cohort: Airways / Oral biological; Skin BIO controls; Control TECH blanks

Arrows show the main data path; enabled analyses may run concurrently. Control decontamination can be bypassed.